

Calibration and Predictive Performance of Continuous-Time and Discrete-Time Stochastic Volatility Models in Option Pricing

Presented by Advisor
ROU-JYUN YU Prof. TING-FU CHEN

National Central University

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Outline

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Introduction & Research Objectives

- **Core Question:** Investigate performance discrepancies between continuous-time stochastic volatility models (Heston) and discrete-time models (GARCH(1,1)) under extreme market conditions.
- **Data:** S&P 500 options. From January to March 2020. Under COVID-19 black swan event.
- **Research Objectives:**
 - ① Evaluate both models' capacity to capture extreme market conditions.
 - ② Analyze the distortion of parameter calibration during high-stress crises.
 - ③ Compare Out-of-Sample (OOS) pricing stability and forecasting accuracy.

Heston Model Dynamics

The Heston model assumes that the stock price S and its variance v follow a system of coupled Stochastic Differential Equations (SDEs) under the risk-neutral measure:

$$dS(t) = \mu S dt + \sqrt{v(t)} S dW_1(t) \quad (1)$$

$$dv(t) = \kappa(\theta - v(t))dt + \sigma\sqrt{v(t)}dW_2(t) \quad (2)$$

Where the parameters are defined as:

- $\kappa > 0$: Speed of mean reversion (rate at which variance returns to θ).
- $\theta > 0$: Long-term mean of variance.
- $\sigma > 0$: Volatility of volatility (controlling the variability of stochastic volatility).
- v_0 : Initial variance level at $t = 0$.

$W_1(t)$, $W_2(t)$ are Wiener processes, and

$$\text{Cov}[W_1(t), W_2(t)] = \rho dt, \quad \rho \in [-1, 1]$$

Negative ρ captures the leverage effect.

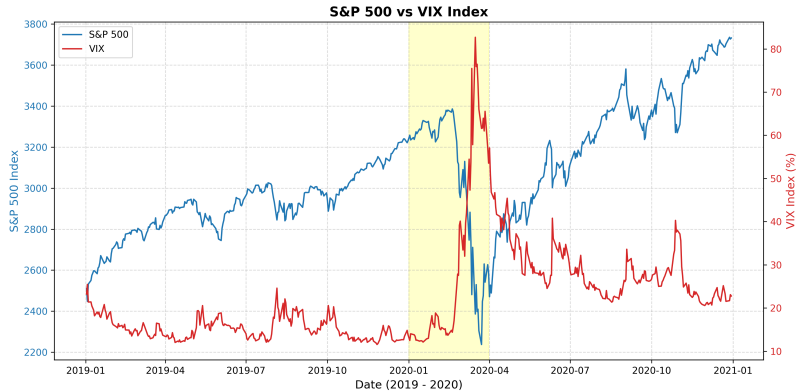


Figure: correlation chart between S&P 500 and VIX index

Heston Model Characteristics Function

$$u = -\frac{1}{2}, \quad b = \kappa + \lambda, \quad a = \kappa\theta, \quad d = \sqrt{(\rho\sigma i\varphi - b)^2 - \sigma^2(2ui\varphi - \varphi^2)}, \quad g = \frac{b - \rho\sigma i\varphi + d}{b - \rho\sigma i\varphi - d}$$

$$C(\tau, \varphi) = ri\varphi\tau + \frac{a}{\sigma^2} \left[(b - \rho\sigma i\varphi + d)\tau - 2 \ln \left(\frac{1 - ge^{d\tau}}{1 - g} \right) \right]$$

$$D(\tau, \varphi) = \frac{b - \rho\sigma i\varphi + d}{\sigma^2} \cdot \frac{1 - e^{d\tau}}{1 - ge^{d\tau}}$$

$$f(x, v, \tau, \varphi) = \exp \left(C(\tau, \varphi) + D(\tau, \varphi)v + i\varphi x \right), \quad x = \ln S$$

Derivation of Heston Characteristics Function

Assume an affine-form solution: $f(x, v, \tau, \varphi) = \exp(C(\tau, \varphi) + D(\tau, \varphi)v + i\varphi x)$

Differentiate and substitute into Heston pde:

$$\frac{\partial f}{\partial \tau} + \left(r - \frac{1}{2}v\right) \frac{\partial f}{\partial x} + \kappa(\theta - v) \frac{\partial f}{\partial v} + \frac{1}{2}v \frac{\partial^2 f}{\partial x^2} + \rho\sigma v \frac{\partial^2 f}{\partial x \partial v} + \frac{1}{2}\sigma^2 v \frac{\partial^2 f}{\partial v^2} = 0 \quad (3)$$

We have

$$\left[\frac{dC}{d\tau} + ri\varphi + \kappa\theta D\right] + v \left[\frac{dD}{d\tau} - \frac{1}{2}i\varphi - \kappa D - \frac{1}{2}\varphi^2 + i\rho\sigma\varphi D + \frac{1}{2}\sigma^2 D^2\right] = 0 \quad (4)$$

Then solve:

$$\frac{dD}{d\tau} = \frac{1}{2}\varphi^2 + \frac{1}{2}i\varphi + (\kappa - i\rho\sigma\varphi)D - \frac{1}{2}\sigma^2 D^2 \quad \text{with } D(0, \varphi) = 0, \quad \frac{dC}{d\tau} = -ri\varphi - \kappa\theta D \quad \text{with } C(0, \varphi) = 0$$

by integral on both side.

Hence, we have the coefficients: $C(\tau, \varphi), D(\tau, \varphi)$

Heston FFT Closed-Form Derivation

$$C(t, K) = e^{-r\tau} \int_k^\infty (e^{s_T} - e^k) Q(s_T) ds_T \quad , \text{ where } s_T = \ln S_T, k = \ln K$$

Since $Q(s_T)$ is unknown, we transform into the Fourier domain via $f_\tau(\omega) = E[e^{i\omega s_T}]$

$$\psi_\tau(\omega) = \mathcal{F}(e^{\alpha k} C(t, k)) = \int_{-\infty}^\infty e^{i\omega k} e^{\alpha k} C(t, k) dk = \frac{e^{-r\tau} f_\tau(\omega - i(\alpha + 1))}{\alpha^2 + \alpha - \omega^2 + i(2\alpha + 1)\omega}, \alpha > 0$$

$$C(t, k) = \frac{e^{-\alpha k}}{2\pi} \int_{-\infty}^\infty e^{-i\omega k} \psi_\tau(\omega) d\omega = \frac{e^{-\alpha k}}{2\pi} \int_{-\infty}^\infty e^{-i\omega k} \frac{e^{-r\tau} f_\tau(\omega - i(\alpha + 1))}{\alpha^2 + \alpha - \omega^2 + i(2\alpha + 1)\omega} d\omega$$

GARCH(p,q)

- Stock dynamics:

$$\log(S(t)) = \log(S(t - \Delta)) + r - \frac{1}{2}h(t) + \sqrt{h(t)}z^*(t) \quad (3)$$

- Variance dynamics:

$$\begin{aligned} h(t) = & \omega + \sum_{i=1}^p \beta_i h(t - i\Delta) + \sum_{i=2}^q \alpha_i \left(z(t - i\Delta) - \gamma_i \sqrt{h(t - i\Delta)} \right)^2 \\ & + \alpha_1 \left(z^*(t - \Delta) - \gamma_1^* \sqrt{h(t - \Delta)} \right)^2, \end{aligned} \quad (4)$$

where, $z^*(t) = z(t) + \left(\lambda + \frac{1}{2}\right) \sqrt{h(t)}$ and $\gamma_1^* = \gamma_1 + \lambda + \frac{1}{2}$.

- Characteristics function:

$$\begin{aligned} f(\phi) = & S(t)^\phi \exp \left(A(t; T, \phi) + \sum_{i=1}^p B_i(t; T, \phi) h(t + 2\Delta - i\Delta) \right. \\ & \left. + \sum_{i=1}^{q-1} C_i(t; T, \phi) \left(z(t + \Delta - i\Delta) - \gamma_i \sqrt{h(t + \Delta - i\Delta)} \right)^2 \right), \end{aligned} \quad (7)$$

GARCH(1,1) Model Dynamics

Log-returns and conditional variance $h(t)$ are specified as:

$$\log \left(\frac{S(t)}{S(t-\Delta)} \right) = r - \frac{1}{2}h(t) + \sqrt{h(t)}z^*(t) \quad (5)$$

where $z^*(t) \sim N(0, 1)$, and the variance updating mechanism recursive loop satisfies:

$$h(t) = \omega + \beta_1 h(t-\Delta) + \alpha_1 \left(z^*(t-\Delta) - \gamma_1^* \sqrt{h(t-\Delta)} \right)^2 \quad (6)$$

- $\omega, \alpha_1, \beta_1$ are non-negative coefficients subject to the stationarity boundary constraint:
 $\alpha_1 \gamma_1^2 + \beta_1 < 1$.
- $\text{Cov}_{t-\Delta}[h(t+\Delta), \log(S(t))] = -2\alpha_1 \gamma_1 h(t)$.

Given positive α_1 , positive value for γ_1 results in negative correlation between spot returns and variance.

Derivation of GARCH(1,1) Characteristics Function

Assume an affine-form solution:

$$f_t(\phi) = S_t^\phi \exp(A_t + B_t h_{t+1})$$

Apply backward recursion

$$f_t(\phi) = E_t[f_{t+1}(\phi)]$$

with

$$f_{t+1}(\phi) = S_{t+1}^\phi \exp(A_{t+1} + B_{t+1} h_{t+2})$$

Substitute the stock price dynamics and raise to the power ϕ

Substitute $h_{t+2} = \omega + \beta h_{t+1} + \alpha(z_{t+1} - \gamma\sqrt{h_{t+1}})^2$ multiply by B_{t+1} .

Combine all exponential terms ,Apply the Gaussian integral formula

$$E[e^{az^2+bz}] = (1 - 2a)^{-1/2} \exp\left(\frac{b^2}{2(1 - 2a)}\right)$$

Substitute back and collect terms:

- Constant terms $\rightarrow A_t$
- Coefficients of $h_{t+1} \rightarrow B_t$

Hence, We have the recursion coefficients: A_t, B_t

GARCH(1,1) Characteristics Function

$$A(t; T, \phi) = A(t + \Delta; T, \phi) + \phi r + B_1(t + \Delta; T, \phi)\omega - \frac{1}{2} \ln(1 - 2\alpha_1 B_1(t + \Delta; T, \phi))$$

$$B_1(t; T, \phi) = \phi(\lambda + \gamma_1) - \frac{1}{2}\gamma_1^2 + \beta_1 B_1(t + \Delta; T, \phi) + \frac{1/2(\phi - \gamma_1)^2}{1 - 2\alpha_1 B_1(t + \Delta; T, \phi)}$$

these coefficients can be calculated recursively from the terminal conditions:

$$A(T; T, \phi) = 0.$$

$$B_1(T; T, \phi) = 0.$$

Thus we have the generating function:

$$f(\phi) = S(t)^\phi \exp(A(t; T, \phi) + B_1(t; T, \phi)h(t + \Delta))$$

Replace ϕ to $i\phi$ is the characteristics function, can be inverted to get the risk-neutral probabilities.

GARCH(1,1) Closed-Form Derivation

Let $x = \ln(S(T))$, and $p(x)$ be the probability density

Define the moment generating function $f(\phi) = E_t[e^{\phi x}]$, $p^*(x) = \frac{e^x p(x)}{f(1)}$

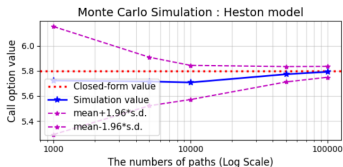
then the mgf of $p^*(x)$ is $\int_{-\infty}^{\infty} e^{\phi x} p^*(x) dx = \frac{1}{f(1)} \int_{-\infty}^{\infty} e^{(\phi+1)x} p(x) dx = \frac{f(\phi+1)}{f(1)}$

$$\begin{aligned} E[\max(e^x - K, 0)] &= \int_{\ln(K)}^{\infty} e^x p(x) dx - K \int_{\ln(K)}^{\infty} p(x) dx = f(1) \int_{\ln(K)}^{\infty} p^*(x) dx - K \int_{\ln(K)}^{\infty} p(x) dx \\ &= f(1) \left(\frac{1}{2} + \frac{1}{\pi} \int_0^{\infty} \operatorname{Re} \left[\frac{K^{-i\phi} f(i\phi + 1)}{i\phi f(1)} \right] d\phi \right) - K \left(\frac{1}{2} + \frac{1}{\pi} \int_0^{\infty} \operatorname{Re} \left[\frac{K^{-i\phi} f(i\phi)}{i\phi} \right] d\phi \right) \end{aligned}$$

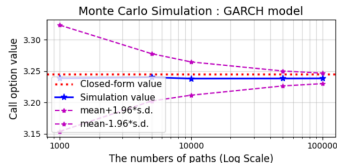
$$\begin{aligned} C &= e^{-r(T-t)} E[\max(e^x - K, 0)] \\ &= \frac{1}{2} S(t) + \frac{e^{-r(T-t)}}{\pi} \int_0^{\infty} \operatorname{Re} \left[\frac{K^{-i\phi} f(i\phi + 1)}{i\phi} \right] d\phi - K e^{-r(T-t)} \left(\frac{1}{2} + \frac{1}{\pi} \int_0^{\infty} \operatorname{Re} \left[\frac{K^{-i\phi} f(i\phi)}{i\phi} \right] d\phi \right) \end{aligned}$$

Closed-Form Verification

Using Monte Carlo simulation through random sampling to construct 95% confidence interval.



Heston



GARCH(1,1)

- Monte Carlo simulation:

Model Dynamics $\rightarrow$ Generate Random Shocks $\rightarrow$ Generate Volatility Paths $\rightarrow$

Generate Stock Price Paths $\rightarrow$ Compute Payoff $\rightarrow$ Average Discounted Payoffs $\rightarrow$ Option Price

Objective Calibration Loss Function

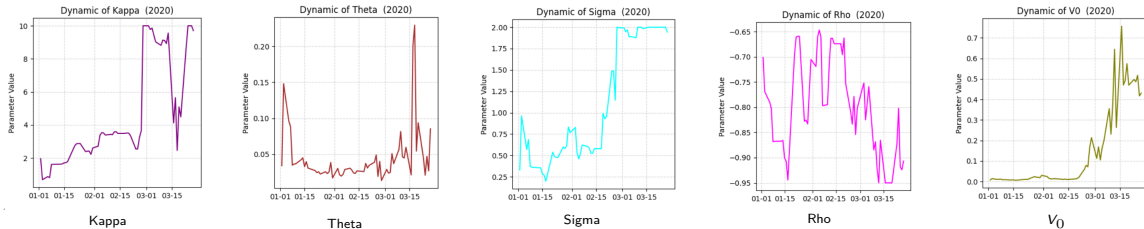
Parameters are calibrated by minimizing pricing errors between model and market prices:

$$\hat{\Theta}_t = \arg \min_{\Theta} \|C_t^{\text{model}}(\Theta) - C_t^{\text{market}}\|^2 \quad (7)$$

- **Heston Target Parameter Vector:** $\Theta_{\text{Heston}} = (\kappa, \theta, \sigma, \rho, v_0)$
- **GARCH(1,1) Target Parameter Vector:** $\Theta_{\text{GARCH}} = (\omega, \alpha, \beta, \gamma, \lambda)$

Calibration Results(Heston)

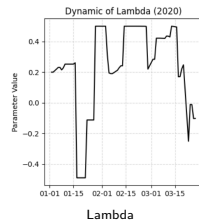
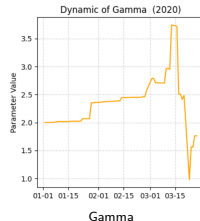
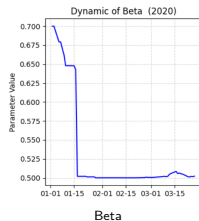
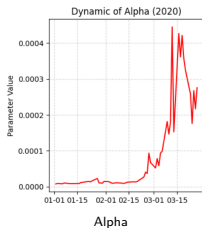
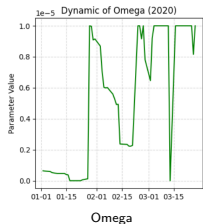
Heston parameters dynamics:



Parameter	Meaning	Average value
κ	Mean reversion speed	4.583878
θ	Long-run variance	0.045379
σ	Volatility of volatility (Vol of Vol)	1.121638
ρ	Correlation between price and variance	-0.803435
ν_0	Initial variance	0.146251

Calibration Results(GARCH(1,1))

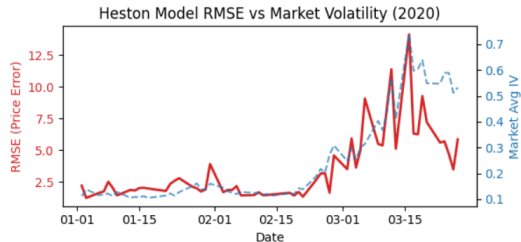
GARCH(1,1) parameters dynamics:



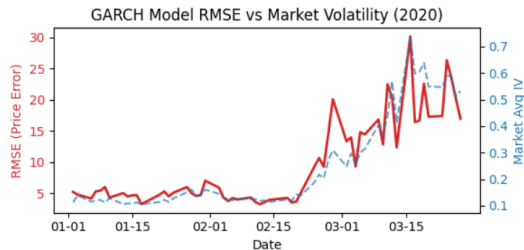
Parameter	Meaning	Average value
ω	Long-run baseline variance	5.68e-6
α	Sensitivity to market shocks	8.723e-5
β	Volatility persistence	0.53162926
γ	leverage effect parameter	2.33507064
λ	Risk premium parameter	0.24473856

RMSE v.s. Market Volatility

High RMSE values coincide with periods of elevated market volatility, with March 2020 representing the peak of the COVID-19 market crisis.

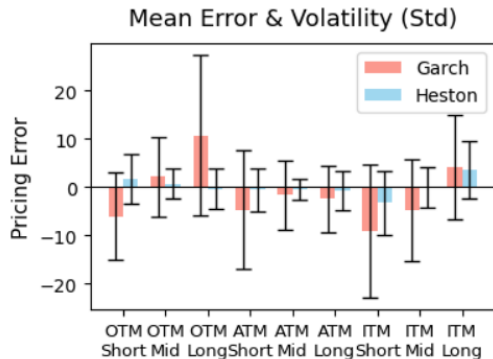


Heston



GARCH(1,1)

Empirical Analysis: In-Sample Results



- Heston model demonstrates stable pricing performance across different maturities. Because its continuous diffusion path can better capture the market volatility.
- GARCH model performs worse for long-maturity OTM options, less stable outside ATM options, because its variance updates in discrete daily intervals.

Heston model provides better pricing performance.

Empirical Analysis: Out-of-Sample Results

Using yesterday's calibrated parameters to estimate today's option prices.

Out-of-Sample RMSE Formula

$$\text{RMSE}_{\text{OOS}} = \frac{1}{\sqrt{N_t}} \left\| \mathbf{C}_t^{\text{model}}(\boldsymbol{\Theta}_{t-1}) - \mathbf{C}_t^{\text{market}} \right\| \quad (8)$$

Heston Model Performance

OOS RMSE = 16.63

Maintains exceptional pricing consistency and stability even through the peak panic of March 2020.

GARCH(1,1) Model Performance

OOS RMSE = 32.64

Error doubles under stress, exposing the latency lag of discrete tracking structures when hit by unexpected jumps.

Core Research Finding

Heston model provides better performance in the extreme market conditions.

The empirical results show that the Heston model provides more stable and consistent option pricing performance during the COVID-19 period. As a continuous-time stochastic volatility model, Heston captures rapid changes in market volatility more effectively than the discrete-time GARCH(1,1) model. Consequently, Heston achieves lower pricing errors and better approximates observed market option prices under extreme market conditions.



Heston, Steven L. (1993).

A Closed-Form Solution for Options with Stochastic Volatility with Applications to Bond and Currency Options.
The Review of Financial Studies, Vol. 6, No. 2, pp. 327–343.



Heston, Steven L., and Saikat Nandi (2000).

A Closed-Form GARCH Option Valuation Model.

Thank You for Your Attention!

Q&A Time